

Project Report: Electric Vehicle Charging Demand Forecasting

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Domain: Data Analytics & Machine Learning

Date: 22/12/2025

1. Abstract

Not so long ago, electric cars started showing up more on roads - but their charge habits? Unpredictable, uneven, often sudden. Power networks now face pressure from these irregular surges in energy draw. To help manage this, a study focused on predicting how much electricity vehicles will need each hour. Forecasts like these could guide decisions about grid stability and where to place new equipment. Instead of relying on just one data source, it pulled together records of past charging events, information on road congestion nearby, and shifts in weather patterns. One method stood out: testing two models' side-by-side, almost like a match between contender and leader. At first, a well-known time-series tool (Facebook's Prophet) set the bar. Then came a different contestant - Random Forest, rooted in decision trees fed with lagged usage trends and movement on streets. That second model didn't just win; it saw sharp jumps in demand others overlooked. Turns out, short-term history plus real-time traffic helped most.

2. Introduction

When urban areas shift toward renewable sources, power networks need foresight - predicting usage before spikes occur. Should electric vehicle charging lack synchronization, minor disruptions might escalate into blackouts. Alternatively, utilities could face pressure to activate costly reserves during high-consumption periods.

Aiming to anticipate total kWh usage over one to two days ahead, this effort focused on crafting a predictive model rooted in observed data. Electric vehicle charging habits - less predictable than standard household electricity use - add complexity, swayed by variables such as weekday rhythms, weather shifts, or congestion during commute peaks. Because these patterns shift unexpectedly, reliable estimates help shape better timing for energy intake. This approach supports reducing strain on infrastructure at high-demand moments, smoothing out supply needs without abrupt spikes.

3. Tools Used

- The project was implemented using a standard Data Science stack:
- Python: Primary programming language for data manipulation and modeling.
- Pandas & NumPy: For time-series manipulation, handling timestamps, and aggregation.
- Using Scikit-Learn to construct the Random Forest Regressor - preprocessing handled through StandardScaler. Tools chosen not for popularity, yet their quiet reliability speaks clearly enough.
- Facebook Prophet: For baseline time-series trend analysis.
- Visualizing data patterns alongside uncertainties often relies on Matplotlib or Seaborn. These tools support initial investigations into datasets while clarifying where errors emerge.

- Tableau: For building the interactive "Grid Operator Dashboard" to visualize heatmaps and traffic correlations.
- A workspace where code runs traceably, its steps recorded alongside results - clarity built into each cell's output.

4. Steps Involved in Building the Project

Phase 1: Data engineering and preprocessing

- Broken down into pieces, the original dataset arrived as three distinct CSVs - each logging events at different rhythms: transaction-level, by the hour, or aggregated per day.
- Breaking down irregular time markers came first - then stitching separate charging episodes together formed an hourly summary of energy demand, measured in kilowatt-hours.
- Bringing together external datasets meant aligning hourly traffic readings with daily weather figures - temperature and precipitation - by extending prior values to fill gaps where timing didn't match. The demand record absorbed these inputs smoothly through time-based propagation.
- From unprocessed timestamps, features emerged - crafted to feed the model effectively
- Time-based cycles handled through sine and cosine adjustments for hour and month values - ensuring that 23 blends smoothly into 0, without abrupt breaks. Periodic patterns remain intact by design, letting adjacent moments stay neighbors in data space.
- Lag Features: Created Lag_1h, Lag_24h, and Rolling_Mean_24h to allow the model to learn from recent history.

Phase 2: Champion-Challenger Modeling

A clear time-based division guided our approach - training drew from the earliest 80%, testing reserved the final fifth, ensuring temporal boundaries stayed intact. One model emerged first; another followed, shaped under similar conditions but distinct in subtle ways

- Model A (The Challenger): Random Forest Regressor. A non-linear ensemble method utilizing all generated features (Lags + Weather + Traffic).
- Model B (The Champion/Baseline): Facebook Prophet. An additive regression model focused on seasonality and trends.

Phase 3: Evaluation & Visualization

- **Metric:** Root Mean Squared Error (RMSE) was used as the primary metric.
- Out of the two, Random Forest delivered notably better results than Prophet. Though Prophet managed to track general movement patterns in usage, its forecasts missed abrupt surges entirely. **Sharp increases - those quick jumps in demand - were simply overlooked. With help from the Lag_1h variable, the Random Forest adjusted almost immediately when such shifts occurred. What stood out was how swiftly it responded to sudden turns in the data flow.**

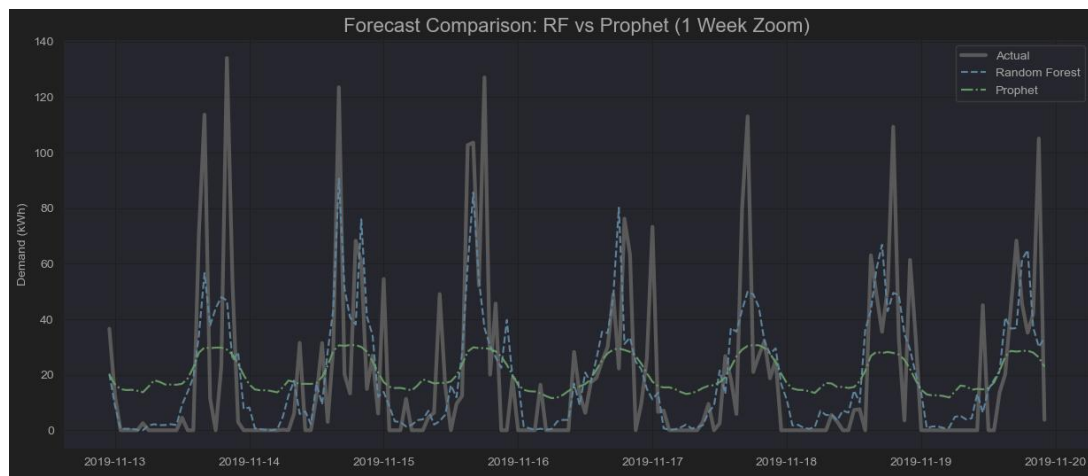


Figure 1: Actual Demand (Gray) vs. Random Forest Prediction (Blue). The RF model closely hugs the actual demand spikes.

5. Conclusion

- This project successfully demonstrated that Machine Learning (Random Forest) is superior to traditional trend-based forecasting (Prophet) for EV charging demand.
- Looking back just one hour turns out to be highly telling when gauging demand. Time of day also plays a central role, shaping patterns in clear ways. Meanwhile, traffic volume contributes less weight - still relevant, yet noticeably behind. These elements stand out, though not equally, in how they help anticipate what comes next.
- A heatmap highlighting anticipated peak times appears in the completed Tableau dashboard, offering grid operators clearer insight. Because of this, planning around high demand becomes more responsive. Raw information gains purpose - shaping decisions that support consistent energy flow. What emerges isn't just visualization, but a functional tool woven into daily operations.

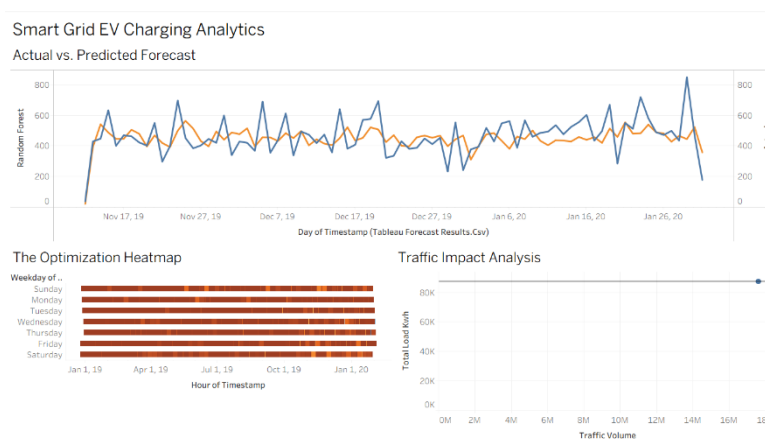


Figure 2: Final Dashboard showing Demand Heatmaps and Traffic Correlations.